LEF MODEL · CHAPTER 7

Chapter 7: The Journey to the Orbit — the Reduced Tangent Law and the Solution of the 5.145° Tilt

Based on Paper 04 — The Orbit (Parts A–E and Appendices A–B)

01 · THE MYSTERY

The mystery: the thirty-year-old problem of the Moon's orbital inclination

For more than thirty years, **the inclination of the Moon's orbit — an angle of 5.145° to the plane of the ecliptic** — was considered one of the most stubborn riddles in planetary science. Why is this angle so hard to explain? Because in all the giant-impact theories, the Moon coalesces from a debris disc swirling around the Earth. Such a disc quickly settles into the Earth's **equatorial plane**, so a moon born from it should begin its life with almost zero tilt — and then late, complicated scenarios are needed (such as encounters with giant planetesimals, orbital resonances or an extreme tilt of the Earth's axis of more than 70°) to tilt the orbit up.

In the LEF model the exact opposite happens: the Moon never coalesced from a flat disc above the equator, but was built on the tip of the pear, which rose to **latitude** $\lambda = 22.5^\circ$ (17,496 km from the Earth's centre, with a 5.77-hour day). From this arise three decisive questions that Paper 04 answers:

1. **What orbit does a body released at latitude 22.5° receive?** Does it enter a stable, receding orbit, or does the Earth's tide pull it inward?
2. **What happens to its orbital plane on the way out** — from a distance of 17,496 km to the Moon's distance today (384,400 km)?

3 **Does the tidal decay law of the inclination turn the 22.5° of the start into exactly the 5.145° of today?**

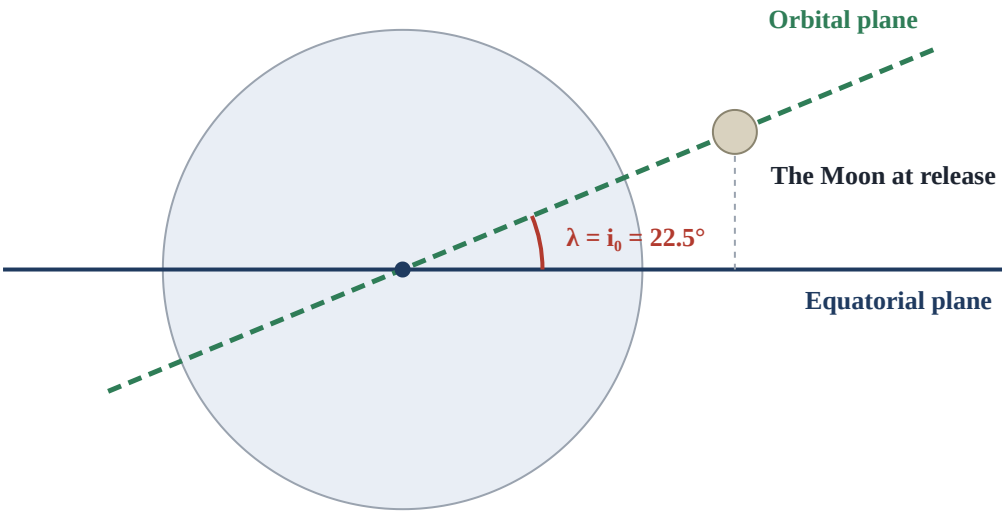
02 · AN IMAGE

An image: the stone released from the raised seat of a merry-go-round

To see with your own eyes why release from latitude 22.5° immediately creates an orbit inclined at 22.5° — with no free parameter and no extra push — imagine a large merry-go-round whose centre lies at the Earth's centre, and whose gravity always pulls exactly toward that centre.

Place a stone on a ring of seats raised above the floor of the merry-go-round (the "latitude" λ). While the merry-go-round turns, the stone moves horizontally eastward, parallel to the floor (the "equatorial plane"). Now let the stone go free! At the moment of release it keeps moving horizontally forward, but gravity pulls it down and inward toward the centre of the merry-go-round. Its free orbit must lie in the only plane that contains both its line of motion and the centre of attraction — **a plane that slopes down diagonally from the stone and passes through the Earth's centre!**

What is the angle between this sloping plane and the floor? Exactly the angle at which the stone is seen from the centre of the merry-go-round above the floor — that is, **the latitude λ itself ($i_0 = |\lambda|$)**! A stone on the floor ($\lambda = 0^\circ$) stays in the plane of the floor; a stone at latitude 22.5° automatically enters an orbit inclined at 22.5°!



The free orbit must pass through the Earth's centre, so its inclination equals the latitude

The kinematic identity: an eastward release at latitude λ gives an orbit whose plane passes through the Earth's centre, with inclination $i_o = |\lambda|$.

03 · SCIENCE AND MODEL

Science and model: three principles that put the orbit on its feet

Parts A and B of Paper 04 show, in three lines of vector algebra, how the detachment conditions of Chapters 2 and 5 turn automatically into a stable, receding orbit:

1. **The kinematic identity ($i_0 = |\lambda|$) and the separation of the three angles (§A.2–A.3):** for a body released with a purely eastward tangential velocity ($v = \omega r(0, \cos\lambda, 0)$), the mass, the distance and the rotation rate cancel completely from the formula, and the exact geometric identity $i_0 = |\lambda| = 22.5^\circ$ results with respect to the early equatorial plane (since an outward radial push at detachment changes the flattening of the ellipse but does not move the orbital plane at all!). At that moment, the 22.5° angle between the Earth's rotation axis and the axis of the Moon's orbit is divided around the total angular-momentum vector ($J = S_\oplus + \mu_r h$) into 3.78° on the Earth's side and 18.72° on the Moon's side.

2. **Why release at the equatorial corotation radius (16,399 km) is impossible (§B.1–B.2):** at latitude $\lambda = 22.5^\circ$, the body's distance from the rotation axis is smaller than its distance from the Earth's centre by a factor $\cos\lambda$. So had the Moon been released at the equatorial corotation distance (16,399 km), its speed would have been too slow: it would have entered an elongated elliptical orbit ($e_0 = \sin^2 22.5^\circ = 0.1464$) at its apogee point, and the Earth's tide would have dragged it inward!

3. **The local corotation line (17,287 km) and the release margin (17,496 km):** at latitude 22.5°, the rotational speed equals the circular orbital speed at a distance of $r = r_c \cos^{-2/3}\lambda = 17,287$ km (x = 1.0542). The separation begins when the centre reaches this line, and the final detachment takes place about 210 km beyond it — at $r_0 = 17,496$ **km** (x = 1.0669, speed ratio B = 1.0367). At this point the Moon enters an almost circular orbit ($e_0 \simeq 0.037$, semi-major axis $a_0 = 18,163$ km, apogee 18,830 km) exactly at its **perigee**, about 7,200 km outside the rigid Roche limit (10,314 km), and the Earth's tide pushes it outward from the very first orbit!

A sharp methodological separation (two reference planes)

The initial angle $i_0 = 22.5^\circ$ is measured relative to the Earth's **early equatorial plane**, while the angle measured today (5.145°) is measured relative to **the plane of the ecliptic** (the plane of the Earth's orbit around the Sun). How does the orbit pass from the equatorial plane to the plane of the ecliptic, and how much of the initial inclination survives the journey outward? Parts C and D of the paper answer this!

04 · THE MECHANISM IN WORDS

The mechanism in words: the Laplace surface, the tangent law and the clock test

As soon as the Moon begins to recede from a distance of 17,496 km toward its present distance (384,400 km), its orbital plane does not stay frozen in place:

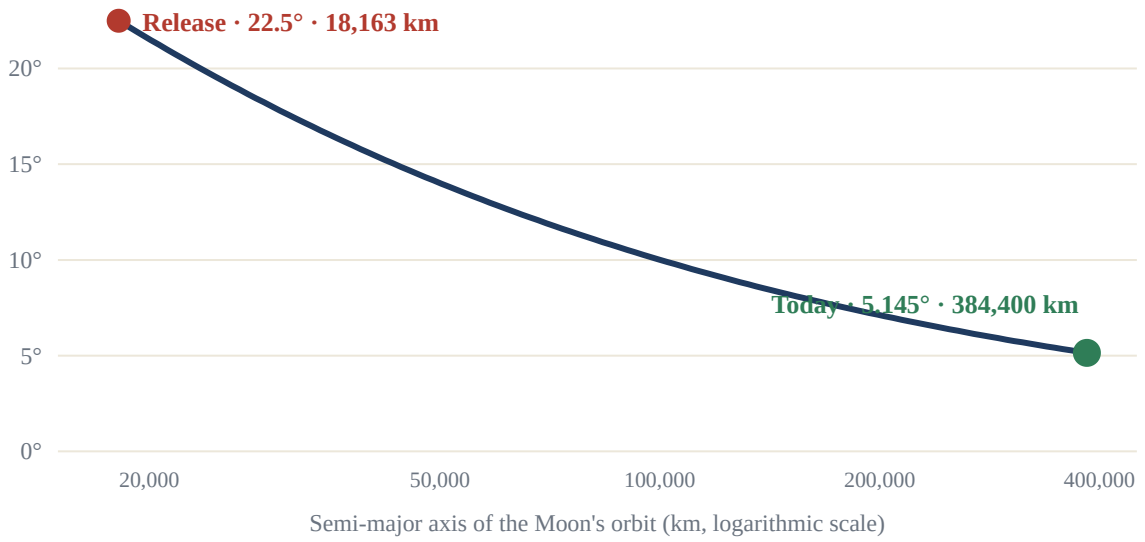
external forces slowly turn it (precession) and the tide damps its inclination. Parts C and D of Paper 04 break this journey into three clear stages:

1. **The passage from the equatorial plane to the plane of the ecliptic — the Laplace surface (§C.1–C.3):** close to the Earth, the equatorial bulge of the spinning Earth (J_2) is much stronger than the Sun's pull, so the Moon's orbital plane precesses around **the equatorial plane** (the Laplace transition radius for a 5.77-hour day is about 17.07 Earth radii, so the release point at 17,496 km lies deep — only about 16.1% — inside the domain controlled by the equator). As the Moon recedes, the control of the equatorial bulge weakens, the Sun's influence grows, and the "guiding plane" (the Laplace surface) bends gradually from the equatorial plane to **the plane of the ecliptic** — while on the way the Moon passes through the freezing of its fossil figure and the Cassini states.

2. **The reduced tangent law — why 22.5° becomes 5.145° (§D.1):** the same tidal forces that push the Moon outward act at the same time on the orbital plane and reduce its inclination. Solving the reduced decay equation shows that the decrease is not according to $\sin i$ (which is only a small-angle approximation), but according to $\tan i \propto a^{-1/2}$: every time the distance grows fourfold, the tangent of the inclination is cut in half!
   When we insert the model's release angle ($i_0 = 22.5^\circ$) and the initial semi-major axis ($a_0 = 18,163$ km) into the equation and run it out to the Moon's present distance ($a_f = 384,400$ km, a 21.16-fold growth in distance), the tangent of the inclination falls by a factor of about 4.6 — and the final angle converges exactly on **5.145°**!

3. **The clock test — why the tidal coefficient $Q_\oplus$ must change with time (§D.2–D.3):** the tangent law links the inclination to the distance, but it does not fix how long the journey takes. When a simple tidal law with a constant friction coefficient ($Q_\oplus = \text{const}$) is applied, it turns out that no single constant value fits both of today's facts together: the value that reproduces today's recession rate ($Q_\oplus \simeq 12 \to 3.70$ cm/yr) brings the Moon to its present distance within only 1.60 billion years; while the value that fits the time elapsed since detachment (3.87 billion years at $Q_\oplus \simeq 29$) gives too slow a

recession rate today (1.5 cm/yr). This confirms the findings of modern research that energy dissipation in the Earth's oceans has changed over history through basin resonances and continental drift!



The reduced tangent law tan i ∝ $a^{-1/2}$: from the release distance to today (a 21.16-fold growth in distance), the inclination falls from 22.5° to 5.145°.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The three central equations of Paper 04 connect the kinematic identity of the moment of release, the initial ellipse of the two masses, and the analytic decay law of the inclination:

$$h = r \times v = \omega r^2(-\sin\lambda\cos\lambda,\ 0,\ \cos^2\lambda) \quad \Longrightarrow \quad \cos i_0 = \frac{h_z}{|h|} = \cos\lambda \ =$$

$$B = \frac{\omega^2 r_0^3 \cos^2\lambda}{K}$$

$$a_0 = \frac{r_0}{2 - B}$$

$$e_0 = |1 - B|$$

$$r_{\mathrm{co}}(\lambda) = r_c \cos^{-2/3}\lambda$$

$$\frac{di}{da} = -\frac{1+\epsilon}{2a}\sin i\cos i \quad \Longrightarrow \quad \tan i(a) = \tan i_0 \left(\frac{a}{a_0}\right)^{-(1+\epsilon)/2}$$

In Appendix A the paper also adds the test of the tidal sign for a constant time lag: the Earth's tidal contribution to the rate of change of the semi-major axis ($\dot{a}$) is positive from the first moment when $\frac{\omega\cos i}{n} > \frac{f_1(e)}{(1-e^2)^{3/2} f_2(e)}$. At the adopted release point (x = 1.0669, $r_0 = 17,496$ km), **1.0769 > 1.0128** holds — so the Moon begins to recede at once, from the very first orbit!

06 · THE SIGNATURE

The signature in reality: no free parameter — release from 22.5° hits 5.145°

The scientific strength of Chapter 7 lies in the fact that the latitude $\lambda = 22.5^\circ$ and the release distance $r_0 = 17,496$ km **were not chosen to fit the Moon's orbit today** — they were fixed in advance from the geometric and mechanical building conditions of Papers 2 and 3 (the position of the pear's tip, the angular-momentum budget, and the Stefan release about 210 km beyond the corotation line).

- **A direct hit on the target:** when these initial conditions ($i_0 = 22.5^\circ, a_0 = 18,163$ km) are inserted into the reduced tangent law ($\epsilon = 0$), the result obtained at a distance of 384,400 km is **5.145°** — identical, to the third decimal place, to the measured inclination of the Moon's orbit to the plane of the ecliptic!
- **Low sensitivity to distance and controlled sensitivity to latitude:** increasing the release distance by 5% (to $a_0 = 19,151$ km) changes the final angle by only 0.14° (to 5.282°), while raising the initial latitude to 25° gives 5.788° —

which shows that the geometry of the pear's tip is what dictates the character of the orbit, while a full vector integration through the Laplace and Cassini transitions is the natural next step of the research.

The lab box: alternative release points, the direction of the tide and the tangent law (Paper 04, Parts B and D)

Release scenario ($T_\oplus = 5.77$ h, $\lambda = 22.5^\circ$)	Release distance (r_0)	Semi-major axis (a_0)	Eccentricity (e_0)	Perigee (r_p)	Direction of the initial tide	Rigid Roche limit test (10,314 km)
On the equatorial corotation line (x = 1.000)	16,399 km	14,304 km	0.1464	12,209 km	Contraction inward (0.753 < 1.196)	Passes (release at apogee)
Exactly on the 22.5° corotation line (x = 1.054)	17,287 km	17,287 km	0.0000	17,287 km	Neutral (exact equality)	Passes (circular orbit)
The adopted release: +210 km beyond the line (x = 1.067)	**17,496 km**	**18,163 km**	**0.0367**	**17,496 km**	**Immediate expansion (1.077 > 1.013)**	**Passes (~ 7,200 km outside)**
8% beyond the equatorial line (x = 1.080)	17,710 km	19,151 km	0.0752	17,710 km	Immediate expansion (1.166 > 1.053)	Passes (release at perigee)
10% beyond the equatorial line (x = 1.100)	18,038 km	20,880 km	0.1361	18,038 km	Immediate expansion	Passes (more elliptical)

Initial conditions in the tangent law ($\tan i \propto a^{-1/2}$)	Latitude (λ)	Semi-major axis (a_0)	Final angle at a distance of 384,400 km	Comparison with observation (5.145°)
The model's release state (+210 km)	**22.5°**	**18,163 km**	**5.145°**	**An exact match to observation!**

Release distance increased by 5% (x = 1.08)	22.5°	19,151 km	5.282°	Low sensitivity to distance (+0.137°)
Sensitivity test at a higher latitude	25.0°	18,163 km	5.788°	Shows that latitude is the controlling parameter

07 · NEXT

The door ahead

We have seen how a body released from the tip of the pear at latitude 22.5° and a distance of 17,496 km enters an almost circular orbit, survives safely outside the rigid Roche limit, begins to recede from the very first orbit, and converges by the tangent law on today's inclination of 5.145°.

Now it is time to look at the face of the Moon itself and listen to the story its rocks, maria and craters tell. **What are the dark "maria" of the Moon — lava lakes that broke out later, or the ancient feeding openings through which the Moon was built and the materials that built its face were ejected?** In the next chapter — **Chapter 8: The Maria of the Moon — Ancient Feeding Openings, Not Late "Lava Lakes"** (from Paper 05 — *The Maria*) — we will go down to the lunar surface!

Chapter 8

The Maria of the Moon — Ancient Feeding Openings, Not Late "Lava Lakes" →

← Chapter 6 · Prologue

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